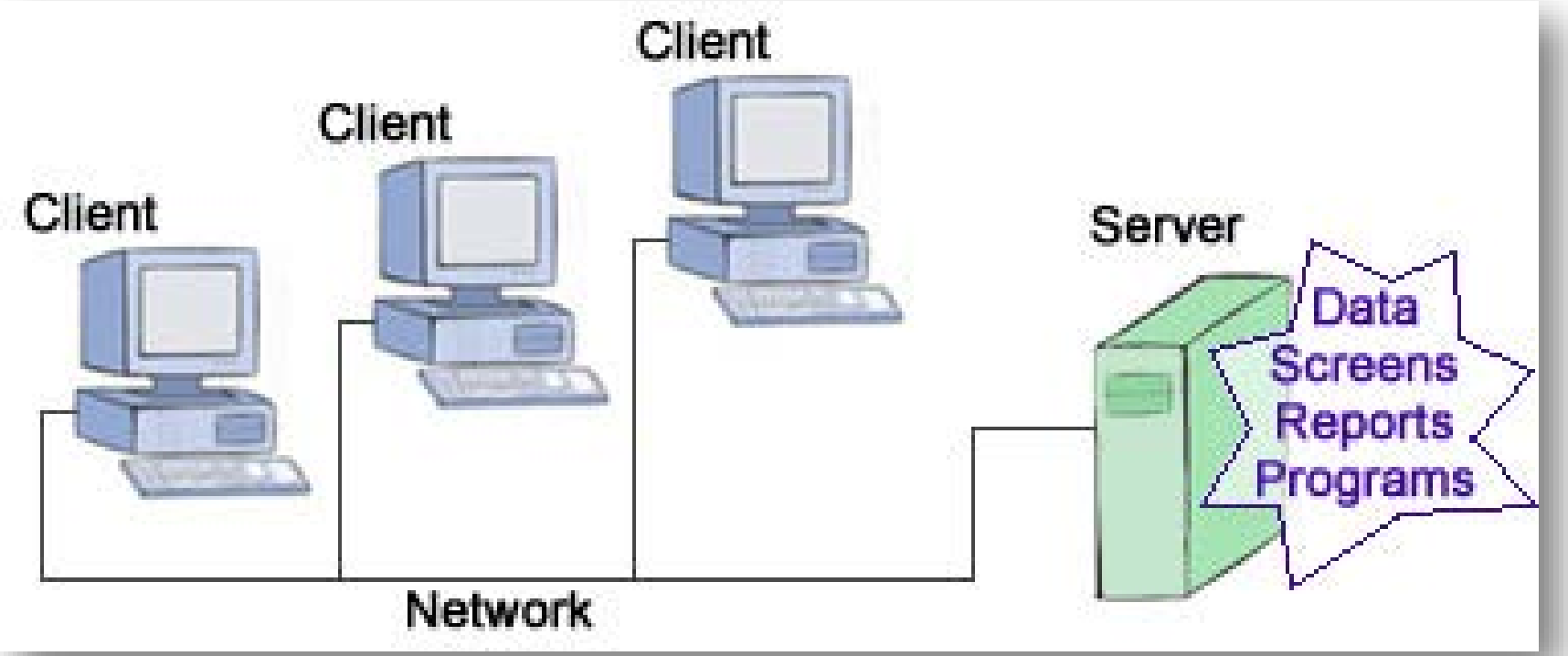
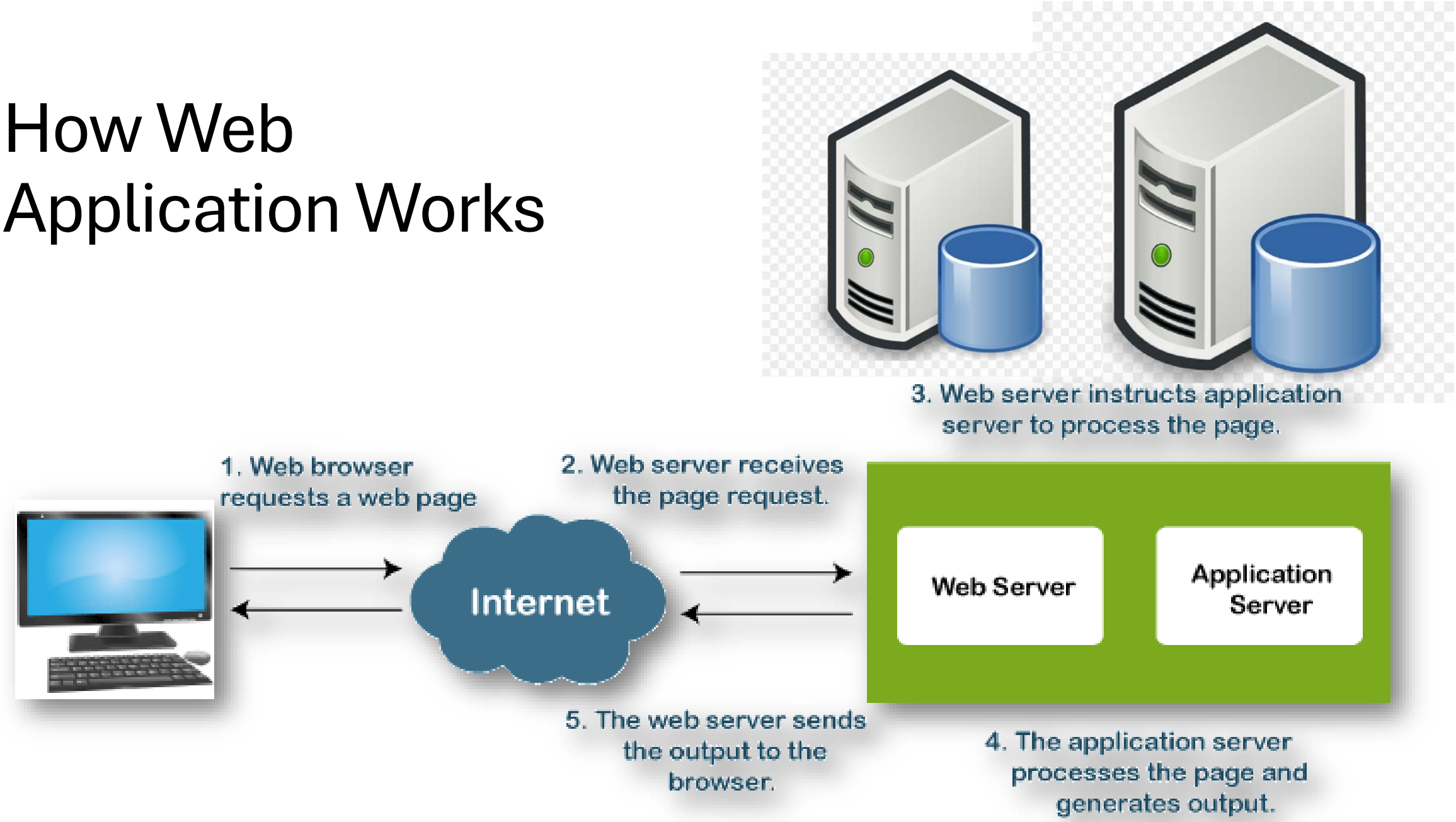
Web Application Dev.

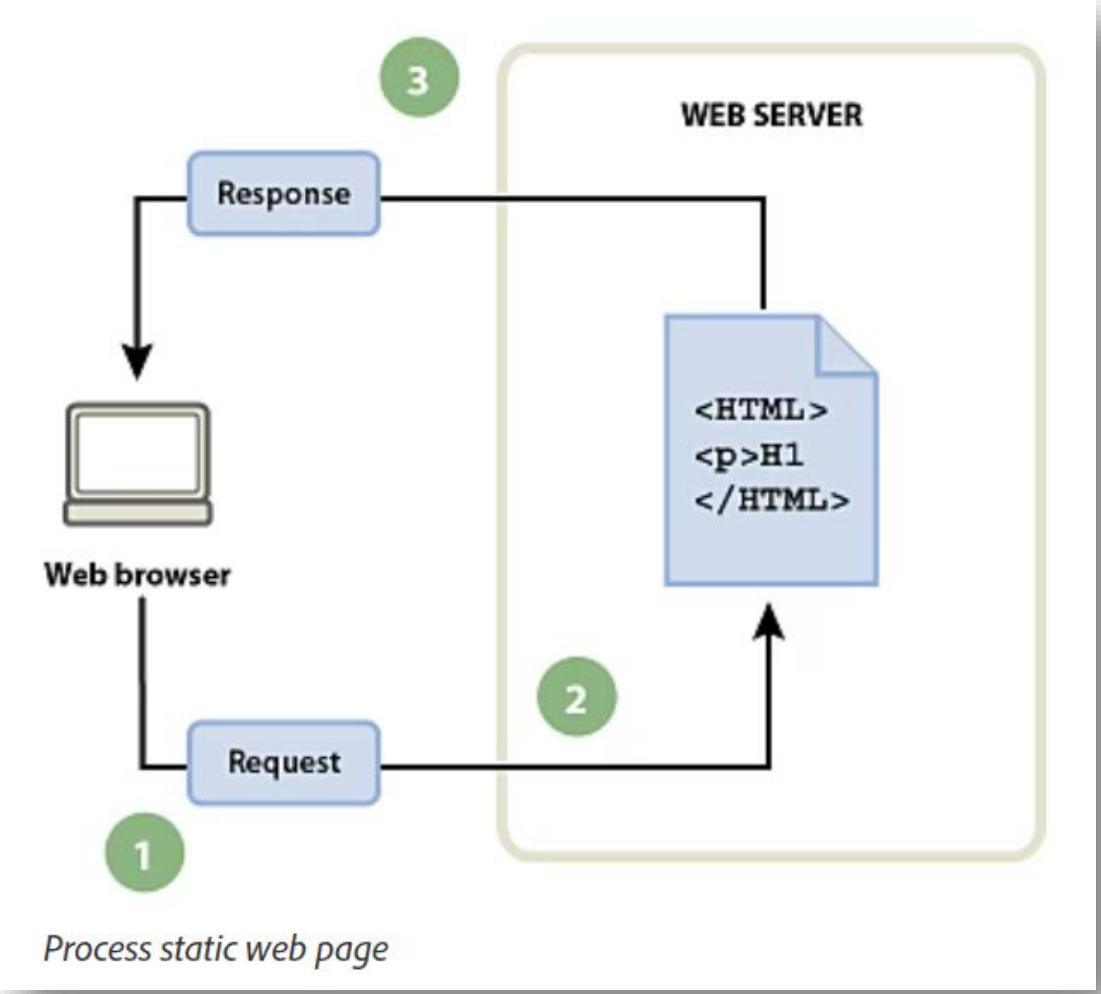
How Client/Server App. Work



How Web Application Works

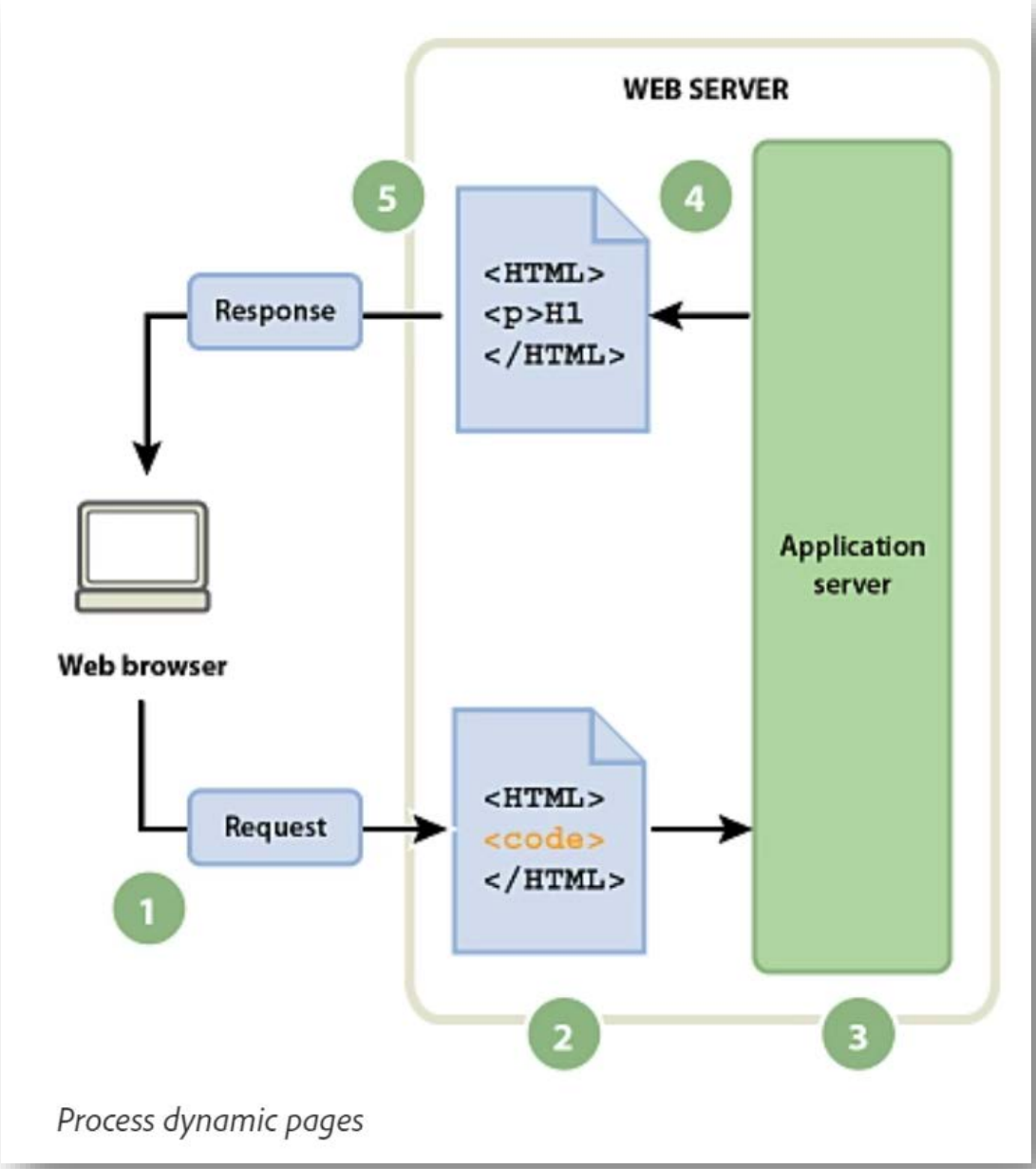


Static Web Page

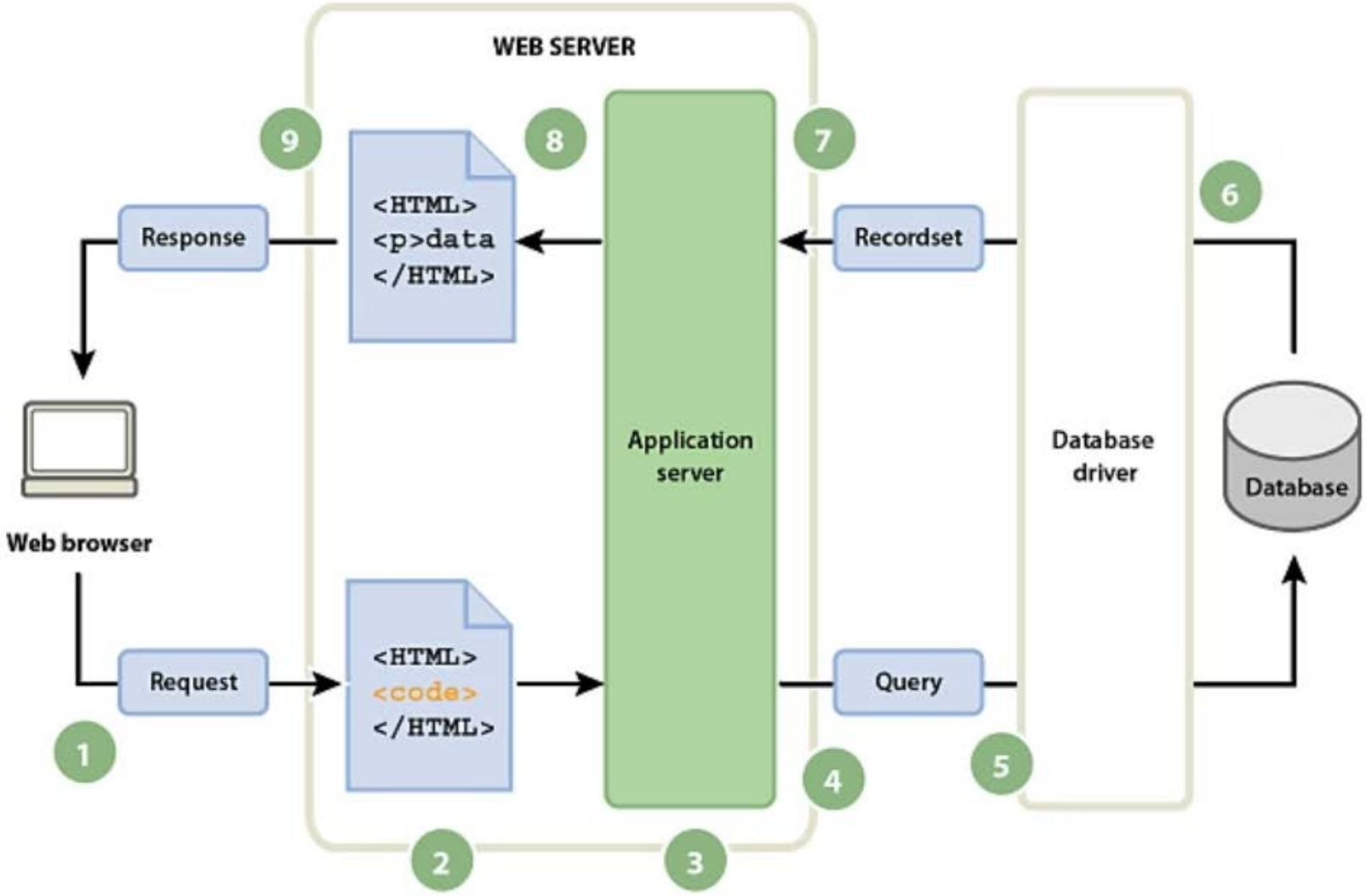


Process static web page

Dynamic Web Page



Process dynamic pages

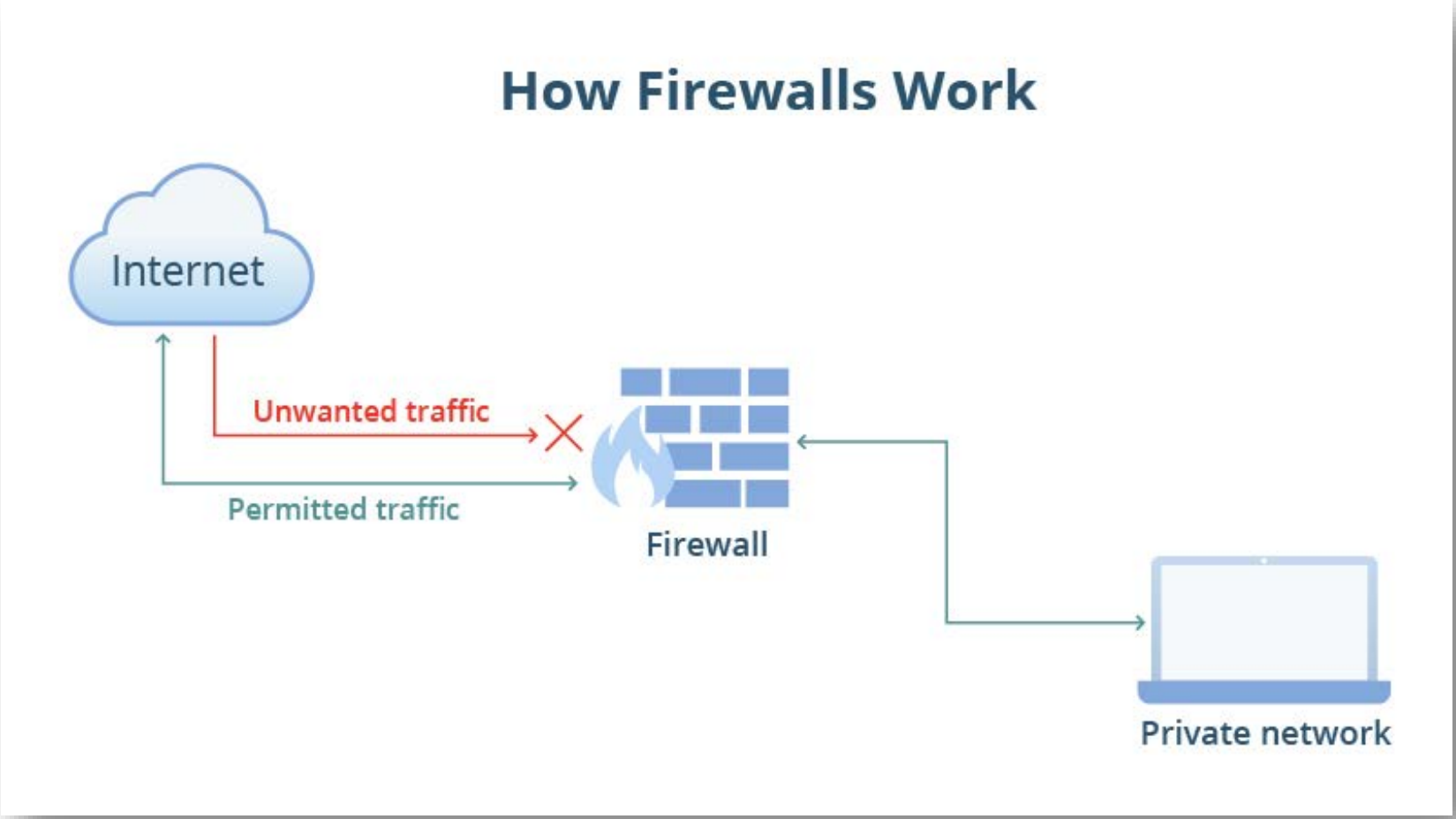


Access a database

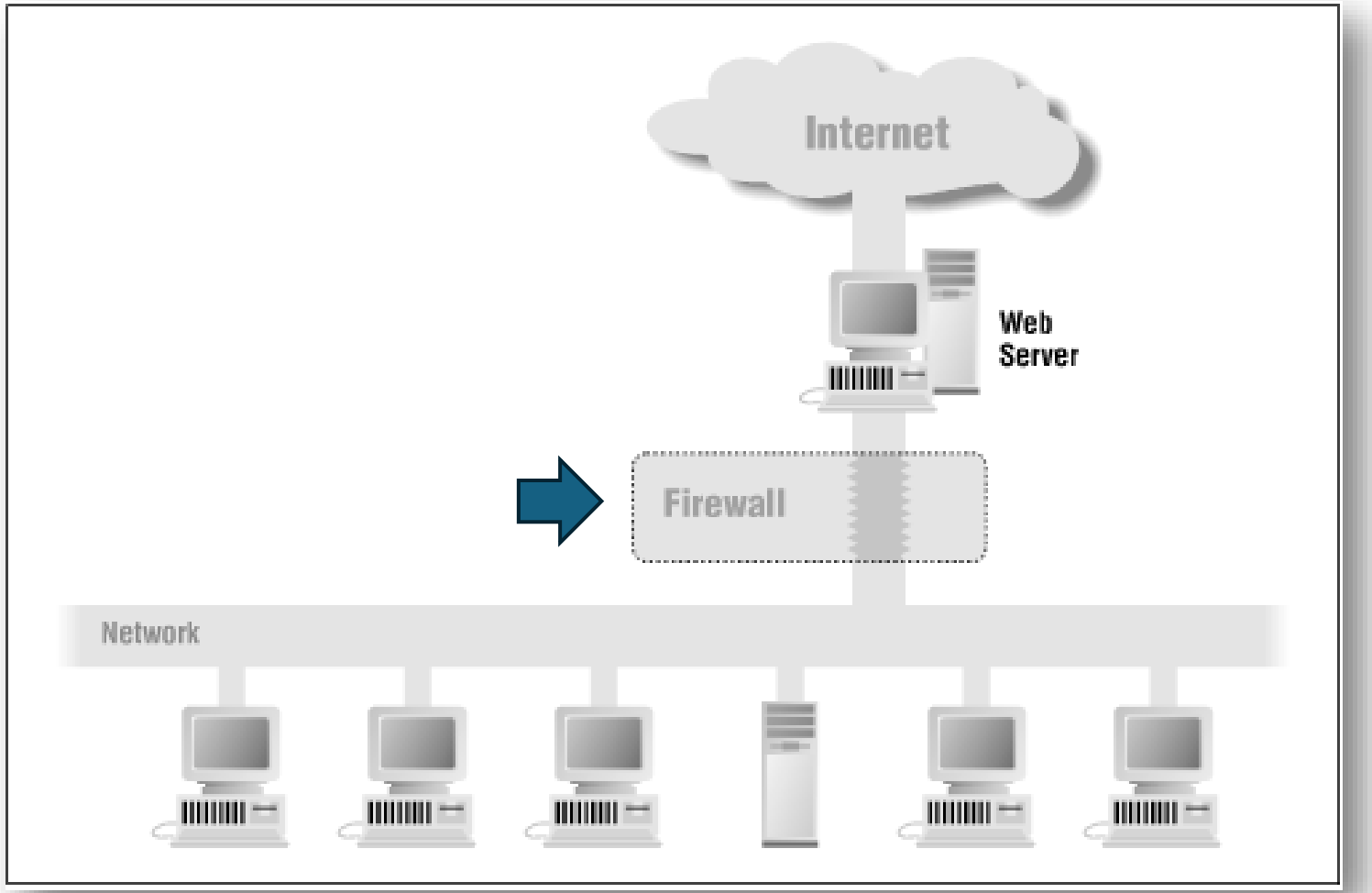
Firewall

Firewall

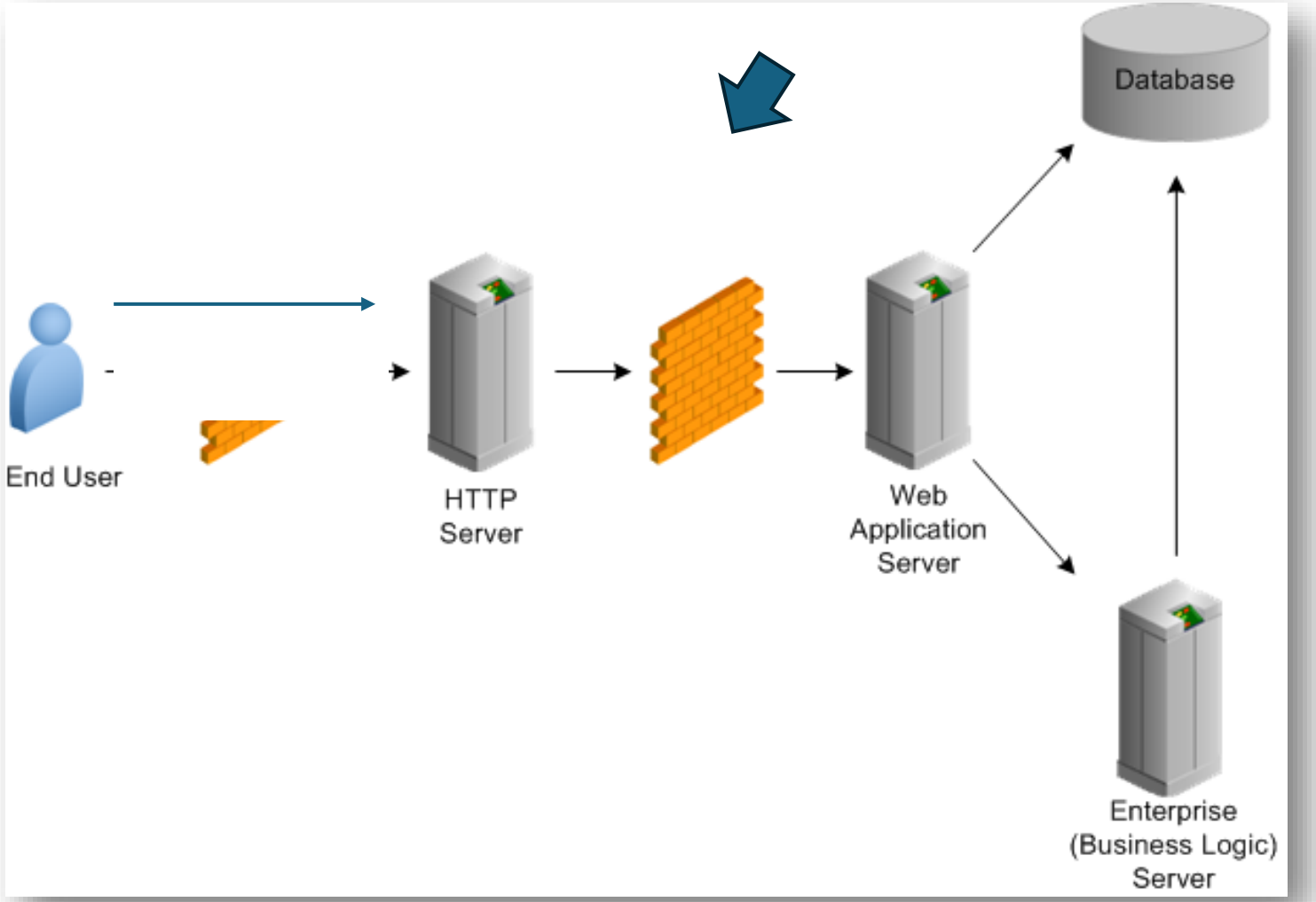
- A firewall is a network security device that prevents unauthorized access to a network
- It inspects incoming and outgoing traffic using a set of security rules to identify and block threats
- A firewall can be physical hardware, digital software, software as a service (SaaS) or a virtual private cloud.



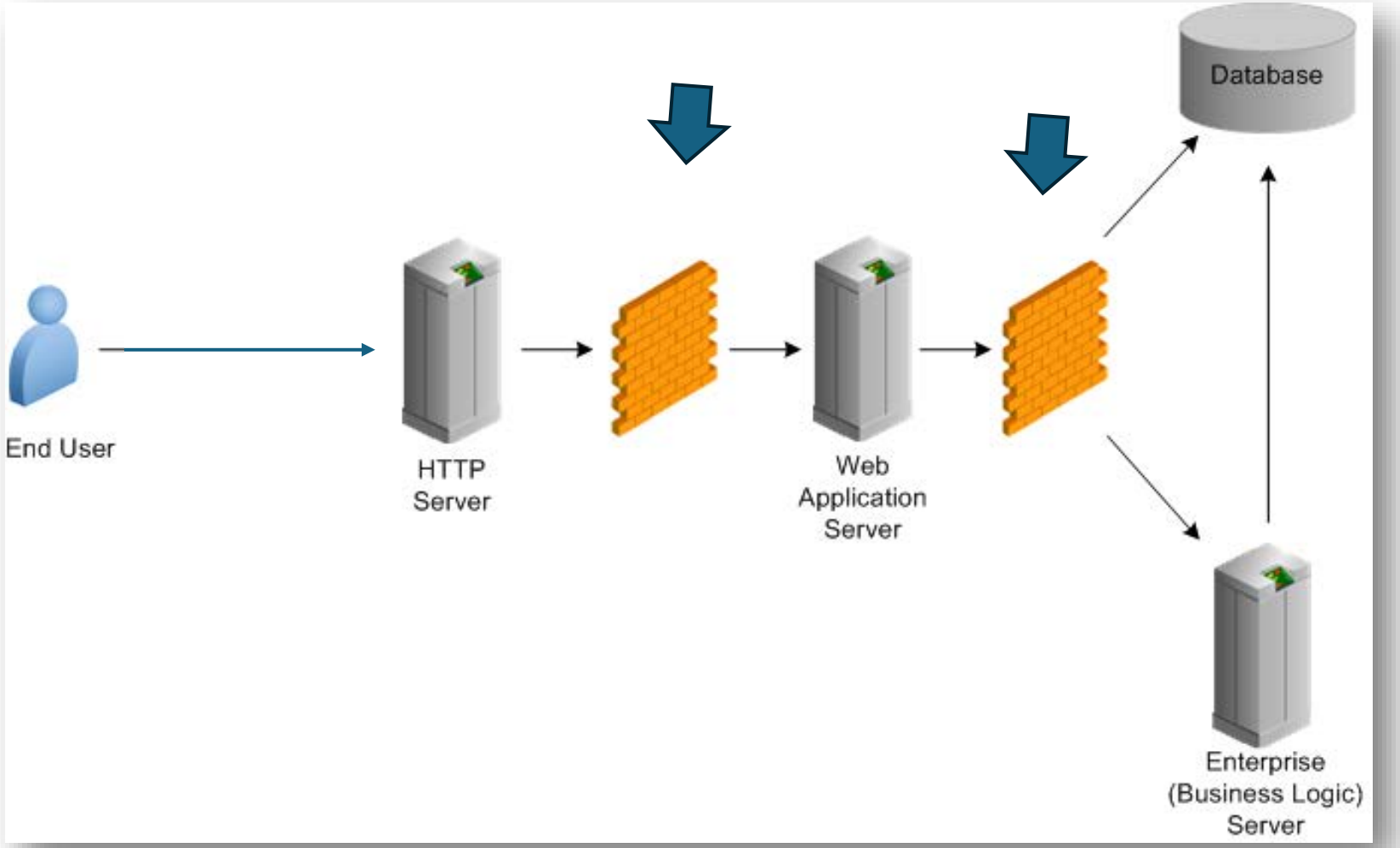
Web Server with Firewall

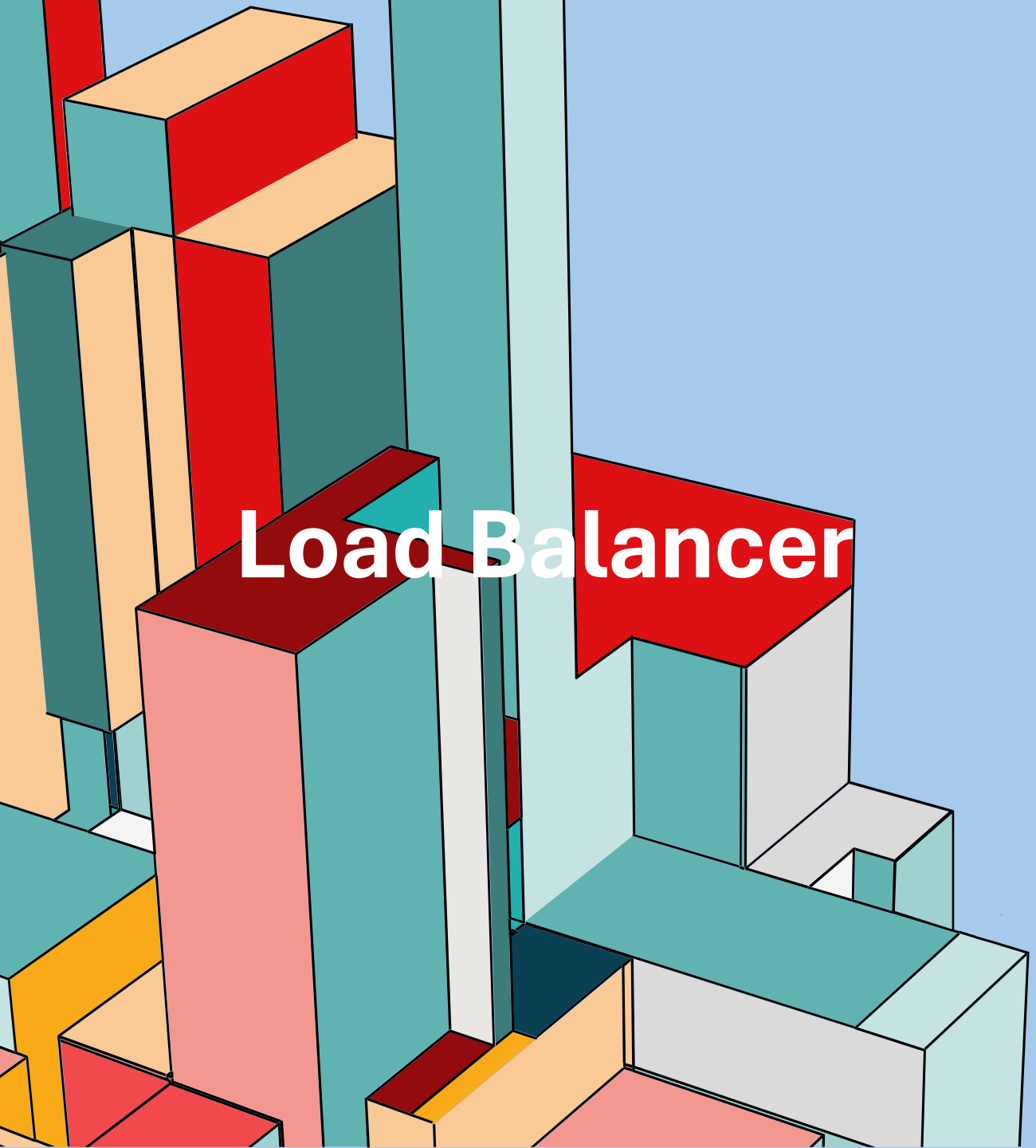


Firewall between Public HTTP Server and App Server



Double Firewall



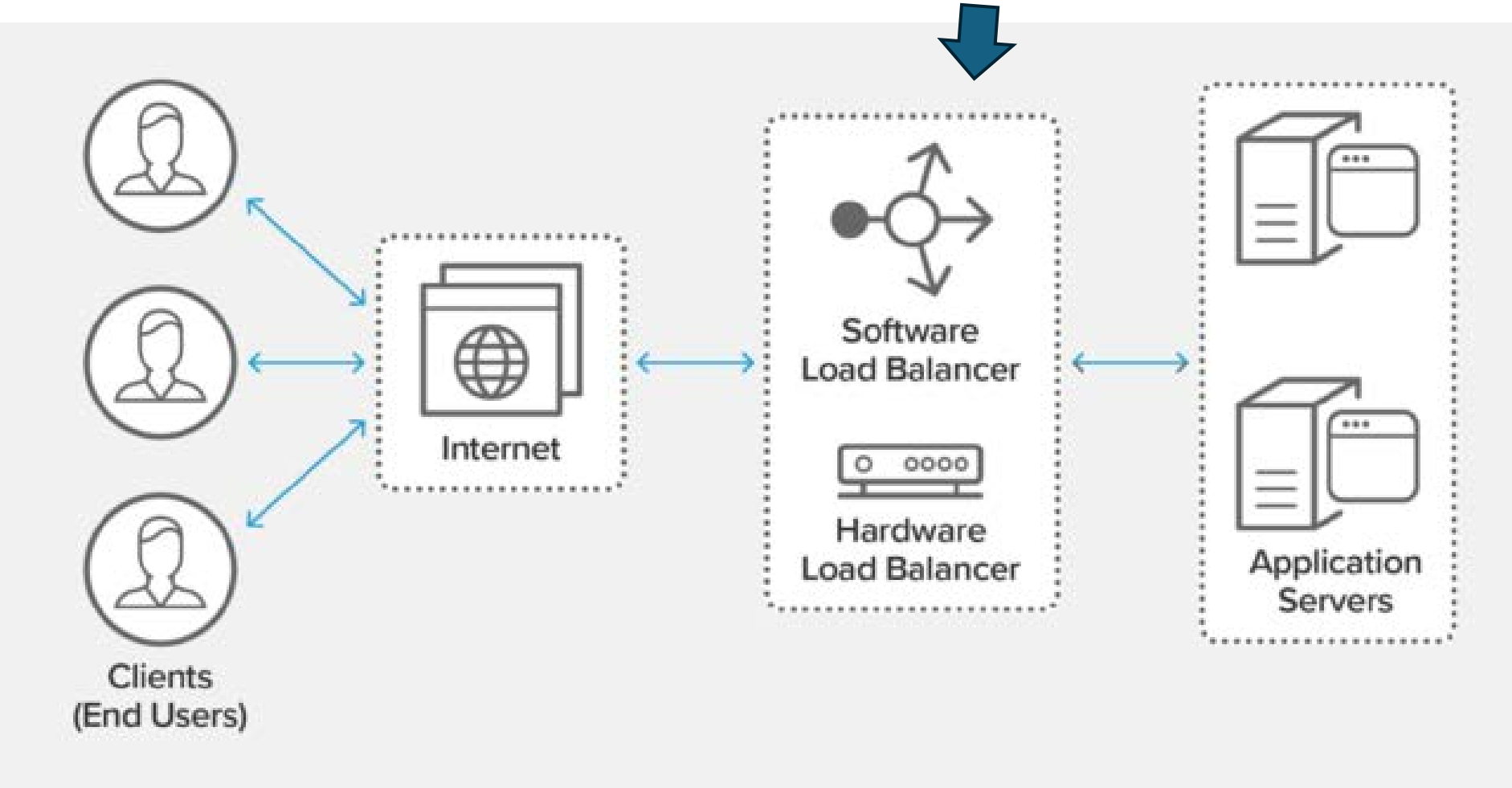


Load Balancer

Load Balancer

- **Load balancing** refers to efficiently distributing incoming network traffic across a group of backend servers, also known as a *server farm* or *server pool*
- Modern high-traffic websites must serve hundreds of thousands of concurrent requests from users or clients
- To cost-effectively scale to meet these high volumes, modern computing best practice generally requires adding more servers
- A **load balancer** acts as the "traffic cop" sitting in front of your servers and routing client requests across all servers capable of fulfilling those requests in a manner that maximizes speed and capacity utilization and ensures that no one server is overworked, which could degrade performance

How to Deploy Load Balancer



https://www.nginx.com/resources/glossary/load-balancing/